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Enhanced Prediction of Type-2 Diabetes Mellitus Using an Integrated Ensemble Framework (IEF)

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***Abstract*—Type 2 Diabetes Mellitus (T2DM) is a pressing global health challenge that significantly reduces life expectancy and diminishes quality of life. Early and accurate diagnosis is critical for avoiding the onset of serious complications. In this study, we proposed an Integrated Ensemble Framework (IEF) that utilizes a Stacking Classifier and a Fusion Model (FM) approach to enhance the accuracy of (T2DM) prediction. Initially, the outputs of two base models, the Support Vector Machine (SVM) and Gradient Boosting Classifier (GBC), are combined using the Stacking Classifier. The combined predictions are then processed through the final decision-making stage, where the Integrated Ensemble Framework is acted based on the Random Forest (RF) algorithm. The results demonstrate that IEF achieves exceptional performance, with an accuracy of 99.09%, precision of 100%, sensitivity of 98.53%, F1 score of 99.26%, and Matthews Correlation Coefficient (MCC) of 98.23%. These findings indicate that the IEF, which integrates outputs from base models, improves overall diagnostic accuracy. This IEF approach leverages the strengths of multiple algorithms, yielding more reliable predictions for T2DM diagnosis. The significant improvement in accuracy validates the potential of hybrid machine learning models in supporting early-stage diagnosis and personalized treatment planning, offering a promising solution for enhancing healthcare outcomes in T2DM patients.**

Keywords—Diabetes Mellitus, Ensemble Machine learning, Stacking Classifier, Fusion Model, Random Forest.

I. INTRODUCTION

The chronic metabolic disease known as diabetes mellitus (DM) is typified by elevated blood sugar levels brought on by the body's inefficient production or use of insulin. There are three categories for it. The immune system's destruction of insulin-producing cells causes type 1 diabetes, sometimes called juvenile or insulin-dependent diabetes, which leads to a total lack of insulin production [1]. For type 2 diabetes, bookkeeping for 90% of diabetes cases has steadily increased globally in recent years [2]. Gestational diabetes mellitus (GDM) occurs during pregnancy, typically in the second or third trimester, and is marked by elevated blood glucose levels and insulin resistance arising from the body's failure to meet the increased insulin demands during pregnancy. Type 2 diabetes makes up about 90-95% of all diabetes cases. In addition to increasing the risk of related disorders such as cardiovascular diseases, stroke, Alzheimer's, cancer, and hypertension, poorly controlled diabetes can cause problems with glucose management. To guarantee accurate prognosis, prompt treatment, and improved patient outcomes, type 2 diabetes must be identified early. Type 2 diabetes is caused mainly by the body's inadequate manufacture of insulin to satisfy its needs, and it is often diagnosed around age 40. Contributing factors include obesity, ageing, and hereditary factors [3]. Additionally, there is a strong correlation between type 2 diabetes and other variables, including age, gender, socioeconomic position, living in an urban or rural area, smoking, drinking alcohol, and eating certain foods [4]. An estimated 537 million individuals worldwide have diabetes [5], while estimates for 2021 indicate that 529 million people have the disease. It is anticipated that the number will increase to 1.31 billion by 2050. Across country boundaries, genders, and age groups, diabetes is a major global cause of death and disability [6]. The cost of diabetes research is estimated to be $760 billion annually [7], with treating diabetes-related complications placing an enormous and escalating financial burden on healthcare systems [8]. Accurately predicting type 2 diabetes for individuals across all age groups is critical because early detection and lifestyle modifications can delay its onset and prevent related health complications [9]. In the past year, data mining and machine learning have emerged as effective tools in the medical field [10]. Healthcare systems may now be automated with machine learning algorithms, increasing the precision of illness prediction. The introduction of distributed computing architectures like Hadoop clusters has facilitated the analysis and storage of vast amounts of data in cloud environments. Hadoop-based clusters were used to test a new machine-learning technique for diabetes prediction [11]. Diabetes has been classified and predicted using machine learning techniques, including a k-nearest neighbour, logistic regression, decision trees, feed-forward neural networks, random forest, support vector machines, artificial neural networks, and naïve Bayes [12]. To enhance the efficiency of detecting type 2 diabetes mellitus (T2DM), we employ an ensemble machine learning approach known as stacking, combined with a new model name replacing the term "IEF". Our first-level base models, SVM and Gradient Boosting Classifier (GBC), undergo rigorous optimization and hyperparameter tuning to boost their performance. At the second level, we apply the stacking ensemble technique, where the predictions from these base models are aggregated. For the final prediction, a model formerly called the "IEF" specifically a Random Forest classifier, is employed to determine whether individuals have type 2 diabetes or not.

The key contributions of this work are summarized as follows:

1. The study introduces the Integrated Ensemble Framework (IEF), which combines the outputs of base

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models (SVM and GBC) through a Stacking Classifier and a Fusion Model (FM) using Random Forest, significantly enhancing T2DM diagnostic accuracy.

2. The IEF achieves a prediction accuracy of 99.09%, surpassing both the base models and the standard Stacking Classifier, demonstrating the effectiveness of model integration in improving T2DM diagnosis.
3. Additionally, the research showcases the perspective of hybrid ML in supporting early-stage T2DM diagnosis and personalized treatment planning, emphasizing the advantages of integrating multiple algorithms.

II. Related Works

A substantial quantity of studies has been carried out with a common goal: accurately identifying early risk factors and symptoms associated with Type 2 Diabetes Mellitus (T2DM), effectively distinguishing at-risk individuals from healthy populations. This classification process aims to aid in analyzing key patterns that contribute to early diagnosis, ultimately assisting in detecting similar risk profiles in new patients. To predict and categorize Type 2 Diabetes in Ethiopia, Ebrahim and Derbew [13] suggested supervised machine learning methods. Thotad et al. [14] uses machine learning approaches for diabetes diagnosis and categorization. A machine learning-based diabetes prediction system is presented by Aizu'bi et al. [15]. An overview of similar studies and their approaches is provided in Table 1.

TABLE I. Related Works Summary

Ref.	Dataset	Model	Accuracy
Maniruzzaman et al. [12]	NHANES	Feature selection (LR) and Classifier (RF)	94.25%
AlZu'bi et al. [15]	PIMA-I	LR	80.2%
Chang et al. [16]	PIMA-I	Nave Bayes	79.13%
Gupta et al. [17]	PIMA-I	Deep Dense Layer Neural Network (DDLNN)	84.42%
Alhalaseh et al. [18]	PIMA-I	Soft Voting (Logistic Regression, Nave Bayes, and Random Forest)	97%
Sivaranjani et al. [19]	PIMA-I	SVM	81.4%
García-Ordás et al. [20]	PIMA-I	CNN	92.31%
Zhou et al. [21]	PIMA-I	K-Means++	98%
Abdulhadi et al. [22]	DDFH-G	ML	82%
Reshan et al. [23]	PIMA-I, DDFH-G, IDPD-I	ANN, LSTM, CNN	98.81%, 97.23%, 94.81%

III. Proposed Methodology

In an effort to progress a reliable ML approach for diabetes detection, this study's methodology incorporates ensemble-based IEF techniques. Several ML techniques are used in the proposed approach, which stacks classifiers with IEF to intensify the accuracy and reliability of diabetes exploration (fig. 1). This section goes into great detail about the process used to create and evaluate the IEF. The key steps that make up this method are explained as follows:

A. Dataset Description

The study's publicly available dataset, SDHS-B, is essential for forecasting diabetes risk and consequences. It comprises 17 elements, including both quantitative and qualitative data, and is based on 520 records from the Sylhet Diabetes Hospital in Sylhet, Bangladesh (SDHS-B). These traits span a wide range of patient-related facts, including joint thrush, polyuria, polydipsia, abrupt weight loss, weakness, polyphagia, blurred vision, itching, irritability, delayed healing, partial paresis, muscle stiffness, baldness, and obesity. The DDFH-G dataset [24] is sourced from Hospital Frankfurt, Germany and includes data on 2000 patients with eight features each. Likewise, there are 768 patient records with eight attributes in the PIMA-I dataset. Although the DDFH-G dataset is larger, both datasets have features in common with those utilized in the investigations by Pradhan et al. and Dwivedi. The characteristics cover a range of health indicators, including skin thickness and the number of pregnancies. Notably, some characteristics, such as blood pressure, skin thickness, insulin, glucose, and BMI, have zero values, which are impractical in real life. The mean of the associated feature column is used to replace these zero values, which are regarded as missing data.

B. Data Preprocessing

Data preparation plays an essential role in the machine learning workflow, transforming raw data into a format that is ready for model training. This process allows learning algorithms to detect significant patterns and relationships within the data. The diabetes prediction system underwent several key preprocessing steps, which included label encoding, normalization, and the partition of data into training and testing sets. Below is a detailed explanation of these procedures:

- Label Encoding: The dataset contains various categorical variables, such as gender (male/female), family history of diabetes (yes/no), and other medical conditions. Since most machine learning algorithms primarily operate on numerical data, categorical variables cannot be directly used as they represent qualitative input. To address this, label encoding was employed. This technique assigns a unique integer to each category in a categorical variable. For example, gender might be encoded as Male = 0 and Female = 1, while family history could be represented as No = 0 and Yes = 1. This transformation is vital as it enables algorithms like SVM, Random Forest, and neural networks to interpret categorical data effectively as part of their input features.
- Normalization: The dataset includes numerical features, such as the patient's age. To standardize the numerical values, particularly the age attribute, min-max scaling was applied for normalization. This technique adjusts the data to a specific range, typically between 0 and 1, using the formula:

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$$X_{norm} = \frac{X - \min(X)}{\max(X) - \min(X)} \quad (1)$$

- Train-Test Split: The dataset was split into a training set with 75% of the data and a testing set with the remaining 25% to evaluate the model's performance and guarantee its capacity to generalize to new data. The machine learning models were trained using the training set to find patterns and connections between the input variables and the desired result. The models were then tested on the testing set to determine how well they performed on data that had never been seen before, mimicking real-world applications. The train-test split is essential for preventing overfitting, ensuring that the model captures generalizable patterns rather than merely memorizing the training data, and validating the model's ability to perform well on new, unknown patient data.

C. *Proposed Integrated Ensemble Framework (IEF)*

In this architecture, we utilize a Stacking Classifier that combines two base models, (SVM) and (GBC), with a Random Forest serving as the IEF. The SVM, equipped with a radial basis function (RBF) kernel, is particularly adept at handling high-dimensional feature spaces and identifying non-linear decision boundaries—making it ideal for distinguishing between diabetic and non-diabetic patients. GBC, as a second base model, enhances performance by sequentially building models that correct the errors of previous ones, which strengthens the model's predictive power. Once the base models (SVM and GBC) have been trained, their predictions are collected and fed into the IEF(RF), which processes them to deliver a final, optimized output. Using RF as the IEF, we leverage its ensemble nature to handle variations in the base model predictions, improving generalization and minimizing overfitting. The hierarchical nature of this approach enables the Random Forest to capture patterns and relationships that the base models may miss, producing a more reliable and accurate classification outcome. This change in architecture—replacing RF as a base model with GBC and using RF as the IEF—improves the system's robustness and overall predictive performance. The step-by-step process of the experiment is depicted in Fig. 1, providing a clear overview of the procedural sequence. The base models run individually, with prediction 1 for SVM and prediction 2 for the GBC classifier. And Stacking classifier combines these base models' output. Then, the output of the stacking classifier would be the input of the Final Integrated Ensemble Framework. We calculate the final evaluation matrices from the output from the Integrated Ensemble Framework.

This IEF approach leverages the strengths of multiple algorithms, yielding more reliable predictions for T2DM diagnosis. The significant improvement in accuracy validates the potential of hybrid machine learning models in supporting early-stage diagnosis and personalized treatment planning, offering a promising solution for enhancing healthcare outcomes in T2DM patients.

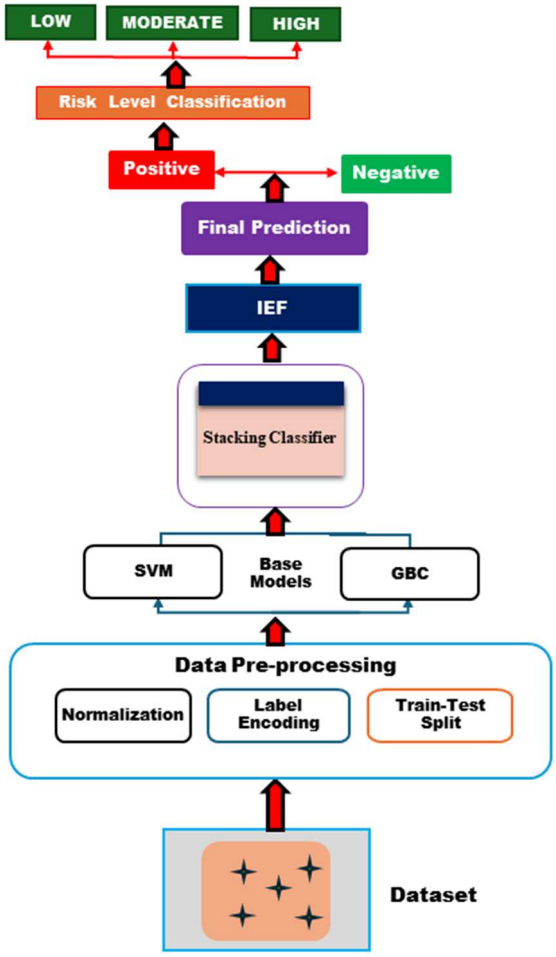

Fig. 1. Proposed Integrated Ensemble Framework (IEF) Architecture

IV. Results and Discussion

Table 2 presents a comparative analysis of performance metrics for the base models (SVM and GBC), the Stacking Classifier, and the IEF based on RF in predicting Type 2 Diabetes Mellitus (T2DM). Regarding accuracy, SVM achieved 98.08%, while GBC performed better with 99%, the Stacking Classifier improved slightly to 99.03%, and the IEF(RF) outperformed all with 99.09%. Precision was perfect (100%) for GBC, the Stacking Classifier, and the IEF, while SVM had a slightly lower precision of 98.44%. Sensitivity (Recall) was similar for SVM and GBC at 98.44% and 98.43%, respectively, with the IEF slightly improving at 98.53%. SVM achieved 97.5% for specificity, while GBC, the Stacking Classifier, and the IEF(RF) all attained perfect specificity (100%). F1 Score, which balances precision and recall, was highest for the IEF at 99.26%, followed closely by the Stacking Classifier at 99.22%, with SVM and GBC at 98.44% and 99.21%, respectively. Finally, Matthew's Correlation Coefficient (MCC), which measures the quality of binary classifications, was highest for the IEF at 98.23%, with the Stacking Classifier slightly lower at 97.99%, and SVM and GBC at 95.94% and 98%, respectively. This table highlights the superior performance of the IEF in this stacking architecture.

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TABLE II. PERFORMANCE EVALUATION OF THE PROPOSED IEF USING THE SDHS-B DATASET.

Evaluation Indexes	Base model		Stacking Classifier	IEF
	SVM	GBC		
Accuracy	98.08	99	99.03	99.09
Precision	98.44	100	100	100
Sensitivity	98.44	98.43	98.43	98.53
Specificity	97.5	100	100	100
F1 Score	98.44	99.21	99.22	99.26
MCC	95.94	98	97.99	98.23

The experimental results depicted in fig.2 compare the performance of various models (SVM, GBC, Stacking Classifier, and IEF) across three different datasets:

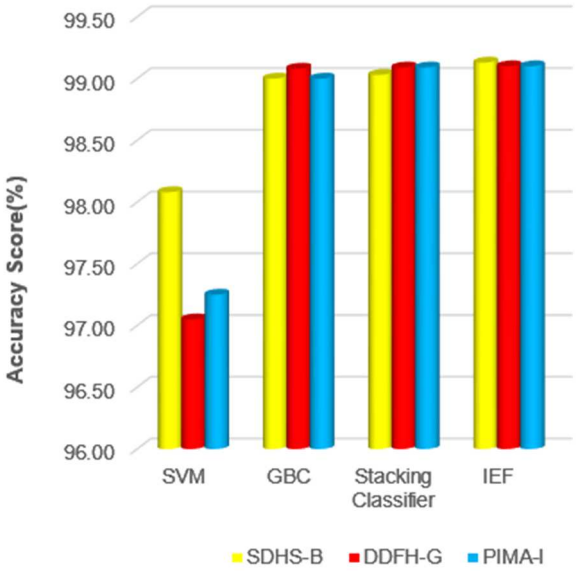

Fig. 2. Performance visualization of the base model, stacking Classifier and IEF using SDHS-B, DDFH-G and PIMA-I datasets.

SDHS-B, DDFH-G, and PIMA-I. For the SDHS-B dataset, SVM achieved an accuracy of 98.08%, while GBC and the Stacking Classifier performed better at 99%, with the IEF (Random Forest) showing the highest accuracy at 99.09%. On the DDFH-G dataset, the SVM performed slightly lower at 97.05%, while GBC achieved 99.08%. The Stacking Classifier and IEF produced similar results, with 99.09% and 99.10% accuracy, respectively. For the PIMA-I dataset, SVM had an accuracy of 97.25%, GBC reached 99%, and the Stacking Classifier and IEF achieved higher performance, each with an accuracy of 99.09% and 99.10%, respectively. Across all datasets, the IEF consistently produced the highest accuracy, demonstrating its strong predictive power.

The ROC curves in Fig. 3 illustrate the comparative performance of the different models used in the study—SVM, Gradient Boosting Classifier (GBC), Stacking Classifier, and the IEF (Random Forest). For every model, the graphs show the true positive rate (sensitivity) about the false positive rate. The SVM model demonstrates strong performance, distinguishing between positive and negative classes. Similarly, the GBC model also exhibits high accuracy with a steep rise in the curve, indicating its strong classification ability. The Stacking Classifier combines the outputs of the base models (SVM and GBC), showing slightly improved performance, suggesting better overall predictive accuracy. Lastly, the IEF performs exceptionally well, with a curve that closely follows the ideal classification line, indicating minimal false positives and strong true positive predictions. This highlights the enhanced performance and robustness of the IEF compared to individual base models.

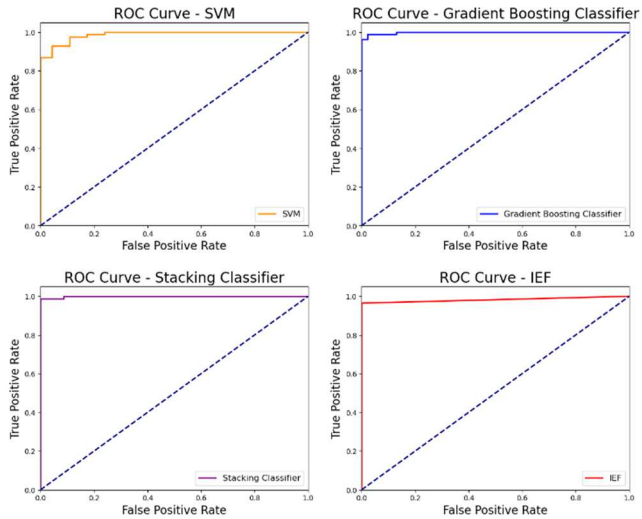

Fig. 3. ROC curve for SVM, GBC, Stacking Classifier, and IEF using SDHS-B dataset.

The learning curves shown in Fig. 4 for multiple models—SVM, GBC, Stacking Classifier, and IEF (RF)—demonstrate how the accuracy scores evolve as the training size increases. Initially, all models show steady improvement with increasing data. The IEF outperforms the other models, achieving the highest accuracy across all training sizes. The Stacking Classifier also performs well, closely following the IEF, especially as more data is provided. IEF(RF) shows strong performance with significant improvement early on, while the SVM has a relatively slower increase in accuracy but improves as the training size grows. The tighter bounds indicate more stable and reliable performance as the models are trained on larger datasets. Overall, the IEF outperform the individual base models, emphasizing the advantage of combining classifiers for enhanced accuracy.

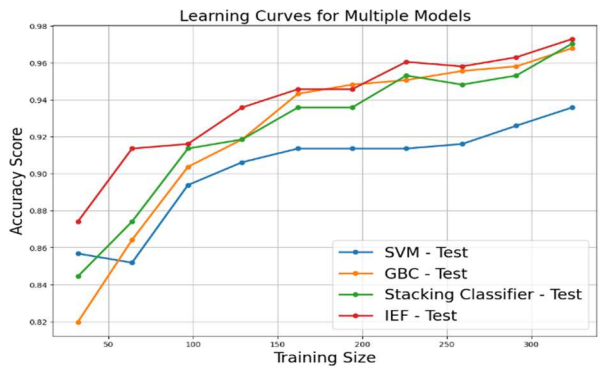

Fig. 4. Learning curve for SVM, GBC, Stacking Classifier, and IEF using SDHS-B dataset.

The proposed IEF (RF) outperformed all other models across multiple datasets. It achieved an impressive 99.03% accuracy

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on the SDHS-B dataset, surpassing previous models applied to the same data.

TABLE III. COMPARING THE PROPOSED IEF WITH STATE-OF-THE-ART APPROACHES.

Ref.	Dataset	Model	Accuracy
[3]	SDHS-B	RF	98%
[12]	NHANES	Feature selection (LR) and Classifier (RF)	94.25%
[13]	DHIS database public hospitals	RF	93.8%
[15]	PIMA-I	LR	80.2%
[16]	PIMA-I	Nave Bayes	79.13%
[17]	PIMA-I	Deep Dense Layer Neural Network (DDLNN)	84.42%
[18]	PIMA-I	Soft Voting (Logistic Regression, Nave Bayes, and Random Forest)	97%
[19]	PIMA-I	SVM	81.4%
[20]	PIMA-I	CNN	92.31%
[21]	PIMA-I	K-Means++	98%
[22]	DDFH-G	ML	82%
[23]	PIMA-I, DDFH-G, IDPD-I	ANN, LSTM, CNN	98.81%, 97.23%, 94.81%
Proposed Model	**SDHS-B**	**IEF**	**99.09%**
	PIMA-I	**IEF**	**99.11%**
	DDFH-G	**IEF**	**99.10%**

Moreover, on the PIMA-I and DDFH-G datasets, the IEF achieved consistent performance with 99.10% accuracy, demonstrating its robustness and effectiveness in different data environments. This highlights that the IEF not only excels in individual dataset performance but also generalizes well across varied datasets, offering a reliable and comprehensive solution for diabetes risk prediction. These results further confirm the IEF's superior accuracy and reliability compared to traditional and machine learning approaches from prior research.

V. CONCLUSION AND FUTURE WORKS

With a high accuracy of 99.11%, our research uses machine learning algorithms—specifically, sophisticated stacking techniques and IEF—to identify Type 2 Diabetes Mellitus (T2DM) early on. Preventing the long-term consequences of diabetes requires early identification, and our method demonstrates how machine learning can be used to develop easily accessible and reasonably priced diagnostic tools.

Future research should concentrate on testing the model with larger, more varied datasets and investigating how variables like body size, height, and BMI may improve prediction accuracy, even if our study shows encouraging findings. Furthermore, the model's capacity to monitor and forecast the emergence of diabetes over time will be enhanced by the integration of longitudinal and time-series data, which will aid in the creation of increasingly sophisticated predictive tools.

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